

System Documentation

Electronics

BlueLine

with Multifunction Display MFD

for Series 4000 FPP

ADEC

Application: Marine

Functional Description

Operating Instructions

Workshop Manual

Installation and

Commissioning Instructions

E532273/02E



Power. Passion. Partnership.

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Table of Contents

1	Safety			
1.1	Important requirements for all products	6		
1.2	Personnel and organizational requirements	7		
1.3	Safety requirements for startup and operation	8		
1.4	Safety requirements for maintenance and repair work	9		
1.5	Fire prevention and environmental protection, fluids and lubricants, auxiliary materials	12		
1.6	Standards for safety notices in the text	14		
2	General Information			
2.1	Engine side and cylinder designations	15		
3	Functional Description			
3.1	Product Summary	16		
3.1.1	BlueLine system - Overview	16		
3.1.2	BlueLine subsystems - Use	19		
3.2	Subsystems	20		
3.2.1	Remote Control System RCS-5	20		
3.2.1.1	RCS-5 BlueLine - Overview	20		
3.2.1.2	RCS-5 BlueLine - Use of devices	22		
3.2.1.3	RCS-5 BlueLine - Main control stand 1	26		
3.2.1.4	RCS-5 BlueLine - Main control stand 2	29		
3.2.1.5	RCS-5 BlueLine - Slave control stand	30		
3.2.2	Monitoring and Control System MCS-5	31		
3.2.2.1	MCS-5 BlueLine - Overview	31		
3.2.2.2	MCS-5 BlueLine - Use of the devices	34		
3.2.2.3	RS422 PIM 4 serial interface as per MTU specifications	37		
3.2.2.4	MCS-5 BlueLine - Main control stand	42		
3.2.2.5	MCS-5 BlueLine - Slave control stand	45		
3.2.3	Engine Control System ECS-5	46		
3.2.3.1	ECS-5 BlueLine - Overview	46		
3.2.3.2	ECS-5 BlueLine - Use of devices	51		
4	Operating Instructions			
4.1	Monitoring and Control System MCS-5	56		
4.1.1	Controls and Displays	56		
4.1.1.1	MCS-5 - Controls and displays	56		
4.1.1.2	LOP display - Overview of screen pages	60		
4.1.1.3	Display MFD - Overview of screen pages	70		
4.1.2	Operation	81		
4.1.2.1	Overall BlueLine system - Switching on	81		
4.1.2.2	Brightness of indicators/instruments - Adjustment	82		
4.1.2.3	Engine - Start	83		
4.1.2.4	Override function - Activation	84		
4.1.2.5	Override function - Deactivation	85		
4.1.2.6	Stopping the engine	86		
4.1.2.7	Emergency engine shutdown	87		
4.1.2.8	Overall system - Switching off	88		
4.1.2.9	PPC - Controls and indicators	89		
4.1.2.10	Oil priming pump control unit PPC - Check function	90		
4.2	Remote Control System RCS-5	91		
4.2.1	Controls and Displays (Standard Version)	91		
4.2.1.1	Command unit for one shaft	91		
4.2.1.2	Command unit for two shafts	93		
4.2.1.3	Command unit for three shafts	95		
4.2.1.4	Command unit for four shafts	97		
4.2.2	Controls and Displays (Special Version)	99		
4.2.2.1	Operating station and rotary encoder modules	99		
4.2.3	Operation	101		
4.2.3.1	Initial command - Allocation	101		
4.2.3.2	Command transfer	102		
4.2.3.3	Operation	103		
4.2.3.4	Engine speed - Change without engagement	104		
4.2.3.5	Single Control Lever mode (SCL) - Activation	105		
4.2.3.6	Single Control Lever mode (SCL) - Deactivation	106		
4.2.3.7	Trolling mode - Activation	108		
4.2.3.8	Trolling mode - Deactivation	109		
4.3	Engine Control System ECS-5	110		
4.3.1	Controls and Displays	110		
4.3.1.1	Local Operating Panel LOP	110		
4.3.1.2	Local Operating Station LOS and Local Operating Panel LOP - Controls and displays	111		
4.3.2	Normal Operation: LOS or LOP with Display	113		
4.3.2.1	Engine - Switching ready for operation	113		
4.3.2.2	Local mode - Activation	114		
4.3.2.3	Engine - Start	115		
4.3.2.4	Local mode - Gearbox engagement/disengagement	116		
4.3.2.5	Local mode - Engine speed adjustment	117		
4.3.2.6	Engine - Shutdown	118		
4.3.2.7	Local mode - Deactivation	119		
4.3.2.8	Engine start - Interlock	120		
4.3.2.9	Engine - Emergency shutdown	121		
4.3.3	Operation during Servicing: LOP without Display	122		
4.3.3.1	Engine - Starting	122		
4.3.3.2	Stopping the engine	123		
4.3.3.3	Engine start - Interlock	124		
4.3.3.4	Engine - Emergency shutdown	125		
4.4	Troubleshooting	126		
4.4.1	Faults in the BlueLine system	126		
4.4.2	Troubleshooting	128		
4.4.3	Alarms - General information	132		

4.4.4	Single-point alarms - Acknowledgment	133	5.1.3.8	Indicator lamp - Replacement	247
4.4.5	Fault messages on the LOP display	134	5.1.3.9	Indicator lamp - Removal	248
4.4.6	Fault indication on MFD display of main control stand	141	5.1.3.10	Indicator lamp - Installation	249
4.4.7	Fault indication on printed circuit board MPU 29	149	5.1.3.11	Indicator lamp - LED replacement	250
4.4.8	Fault messages - Engine Control Unit	154	5.1.3.12	Contactor - Replacement	252
4.4.9	Oil priming pump PPC - Troubleshooting	201	5.1.3.13	LED in pushbutton/switch - Replacement	253
4.5	Task Description	202	5.1.3.14	Display MFD - Replacement	255
4.5.1	Wiring (General) for Engine/Gearbox/Unit	202	5.1.3.15	Battery on MFD display - Replacement	256
4.5.1.1	Engine wiring - Check	202	5.1.3.16	SD card on MFD display - Replacement	258
4.5.2	Control and Display Elements, General	203	5.1.3.17	PIM 4 parameters - Reset	259
4.5.2.1	Indicator lamp - LED replacement	203	5.1.3.18	Fuse in control unit PIM 4 - Replacement	260
			5.1.3.19	Control unit PIM 4 - Replacement	261
			5.1.3.20	Printed circuit board in module cassette - Replacement	263
			5.1.3.21	Fuse on printed circuit board in PIM cassette - Replacement	265
			5.1.3.22	Printed circuit board MPU 29 - Removal	267
			5.1.3.23	Printed circuit board MPU 29 - Installation	268
			5.1.3.24	Printed circuit board MPU 29 - Replacement	269
			5.1.3.25	Data module MEM on MPU 29 - Replacement	270
			5.1.3.26	Data module MEM on MPU 29 - Transfer	271
			5.1.3.27	LOP fuse - Replacement	272
			5.1.3.28	LOP with display front flap - Replacement	273
			5.1.3.29	LOP - Replacement	275
			5.1.3.30	Local Operating Station LOS - Replacement	276
			5.1.3.31	Gear Control Unit GCU - Replacement	277
			5.1.3.32	Priming Pump Controller PPC - Repair	279
			5.1.3.33	Priming Pump Controller PPC - Replacement	281
			5.1.4	Supplementary Technical Information	282
			5.1.4.1	PIM 4 RCS - Internal design	282
			5.1.4.2	Control unit PIM 4 MCS - Internal design	287
			5.1.4.3	Local Operating Panel LOP - Design	293
			5.1.4.4	Gear Control Unit GCU - Design	301
			5.1.4.5	Priming pump control unit PPC design	306
5	Workshop Manual		6	Installation and Commissioning Instructions	
5.1	Task Description	205	6.1	Installation	307
5.1.1	Operating Tests	205	6.1.1	Preparation	307
5.1.1.1	Test mode, BlueLine - Execution	205	6.1.1.1	Cable routes and openings between all installation locations - Check	307
5.1.1.2	Overspeed test	208	6.1.1.2	Cables - Routing between installation locations	308
5.1.1.3	Emergency stop test	209	6.1.2	Mechanical Installation	310
5.1.1.4	Cranking	210	6.1.2.1	PIM 4 module housing - Installation	310
5.1.2	Checks and Settings	211	6.1.2.2	OceanLine analog display instruments - Installation	311
5.1.2.1	LOP-DIS settings - Check	211	6.1.2.3	Horn VDO - Installation	313
5.1.2.2	LOP display DIS - Setting	212	6.1.2.4	Indicator lamp - Installation	314
5.1.2.3	Display MFD settings - Check	213	6.1.2.5	Illuminated pushbuttons/key switches - Installation	315
5.1.2.4	Display MFD - Setting	214	6.1.2.6	GCU - Installation	317
5.1.2.5	PIM 4 node number - Check	216	6.1.2.7	Priming Pump Controller PPC - Installation	318
5.1.2.6	PIM 4 node number - Setting	217	6.1.2.8	Display MFD - Installation	319
5.1.2.7	Serial interface terminator plugs - Check	218	6.1.2.9	LOP - Installation	320
5.1.2.8	Emergency stop shorting plug X18 - Check	219			
5.1.2.9	Jumper configuration of MCS-5 control unit - Check	221			
5.1.2.10	CAN bus connections PIM 4 - Check	222			
5.1.2.11	CAN bus line resistance - Check	224			
5.1.2.12	MPU 29 node address in LOP - Check	226			
5.1.2.13	MPU 29 node address in LOP - Setting	227			
5.1.2.14	Control stand configuration in Local Operating Panel LOP - Check	228			
5.1.2.15	Shaft number setting - Check	230			
5.1.2.16	LOP power supply - Check	231			
5.1.2.17	Jumper configuration of LOP - Check	234			
5.1.2.18	GCU 6 jumper configuration - Check	235			
5.1.2.19	GCU 3 jumper configuration - Check	236			
5.1.2.20	GCU node number - Check	237			
5.1.2.21	Configuration	238			
5.1.3	Repair Work	240			
5.1.3.1	PIM 4 cover - Removal	240			
5.1.3.2	PIM 4 cover - Installation	241			
5.1.3.3	Command unit - Replacement	242			
5.1.3.4	Rotary encoder module - Replacement	243			
5.1.3.5	Operating station - Replacement	244			
5.1.3.6	Analog display instrument VDO OceanLine - Replacement	245			
5.1.3.7	Lamp in analog display instrument VDO OceanLine - Replacement	246			

6.1.2.10	LOS - Installation	322	6.1.4	Data Transmission	382
6.1.2.11	Command unit ROS 11 - Installation	324	6.1.4.1	RS422 - Transmission values	382
6.1.2.12	Command unit ROS 7 / ROS 9 - Installation	326	6.1.5	Technical Data	383
6.1.2.13	Rotary encoder modules ROS 10/P and ROS 10/S - Installation	328	6.1.5.1	Technical data - LOP	383
6.1.2.14	Remote Operating Station ROS 10/T - Installation	329	6.1.5.2	Display MFD - Technical data	384
6.1.2.15	Terminal box X010 - Installation	331	6.1.5.3	ECU 7 - Technical data	386
6.1.3	Electrical Installation	332	6.1.5.4	GCU 3 - Technical data	388
6.1.3.1	HSK cable gland - Preparation	332	6.1.5.5	GCU 6 - Technical data	389
6.1.3.2	Terminal block - Preparation	333	6.1.5.6	PPC - Technical data	390
6.1.3.3	PIM 4 cover - Removal	334	6.1.5.7	Technical data X010	391
6.1.3.4	GCU 3 - Connection	335	6.1.5.8	PIM 4 control unit - Technical data	392
6.1.3.5	GCU - Pin assignment for non-MTU gearbox	337	6.2	Initial Operation	393
6.1.3.6	Display instruments on main control stand - Connection	340	6.2.1	Preparation	393
6.1.3.7	Controls and status indicator lamps on main control stands - Connection	341	6.2.1.1	Checks prior to startup	393
6.1.3.8	Indicator lamps for single-point alarms - Connection	342	6.2.1.2	Operating voltage - First power-up	395
6.1.3.9	Devices on a slave control stand - Connection	343	6.2.1.3	Supply voltage distribution - Check	397
6.1.3.10	PIM 4 MCS control unit - Connection	344	6.2.1.4	Data module MEM on MPU 29 - Installation	400
6.1.3.11	PPC - Connection (single phase)	347	6.2.1.5	SD memory card in display MFD - Installation	401
6.1.3.12	PPC - Connection (three phases)	349	6.2.2	Settings	402
6.1.3.13	Starter and battery-charging generator - Connection	351	6.2.2.1	Configuration	402
6.1.3.14	Engine Control Unit ECU 7 - Connection	356	6.2.2.2	Minidialog - Settings	404
6.1.3.15	Barring tool limit switch - Connection	357	6.2.3	Tests	417
6.1.3.16	Display MFD - Connection	358	6.2.3.1	Initial engine startup at service interface of LOP	417
6.1.3.17	LOP - Connection	360	6.2.3.2	Initial engine startup at the display of Local Operating Panel LOP	419
6.1.3.18	Additional Yard signals (optional) - Connection	363	6.2.3.3	Initial engine start at MCS	421
6.1.3.19	LOS - Connection	365	6.2.3.4	Speed control and clutch setting - Check	423
6.1.3.20	Command units on triple-shaft propulsion systems - Connection	366	6.2.3.5	Command transfer - Check (only when several control stands are installed)	424
6.1.3.21	Command units on propulsion systems with four shafts - Connection	369	6.2.3.6	System reset	425
6.1.3.22	Command unit ROS 7 / ROS 9 - Connection	372	7	Appendix A	
6.1.3.23	Rotary encoder module and remote operating station - Connection	373	7.1	Abbreviations	426
6.1.3.24	PIM 4 RCS control unit - Connection	375	7.2	Conversion tables	431
6.1.3.25	RCS extension module - Installation in PIM 4 RCS	378	7.3	MTU contact persons/service partners	435
6.1.3.26	Terminal box X010 - Connection (battery-charging generator charges starter battery)	379	8	Appendix B	
			8.1	Special Tools	436
			8.2	Index	437

1 Safety

1.1 Important requirements for all products

Nameplate

The product is identified by a nameplate, model designation or serial number which must match the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

General information

This product may pose a risk of injury or damage in the following cases:

- Improper use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications
- Noncompliance with the safety instructions and warning notices

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine management/parameters)
- In compliance with all safety regulations and in accordance with all warning notices in this manual
- With maintenance work performed in accordance with the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use shall be considered non-intended. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Changes or modifications

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

1.2 Personnel and organizational requirements

Organizational measures of the operator

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only.

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

1.3 Safety requirements for startup and operation

Safety requirements for startup

The product must be installed and accepted in accordance with manufacturers' specifications prior to initial startup.

All the requisite regulatory approvals must have been granted and all startup requirements fulfilled prior to initial startup.

Whenever the product is started, always ensure that:

- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- Wearers of cardiac pacemakers or other active medical implants are well clear.
- The operating room is adequately ventilated.
- Exhaust pipework is leak-tight and routed to atmosphere.
- Battery terminals, generator terminals and cables are guarded to preclude accidental contact.
- All personnel is clear of the danger zone represented by moving parts.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems are working properly.

Safety requirements for operation

The operator must be familiar with the controls and displays.

The operator must be aware of the consequences of any operations he/she performs.

During operation, the display instruments and monitoring units must be constantly observed in regard of present operating status, limit value violation and warning or alarm messages.

Malfunctions and emergency stop

Emergency procedures, in particular emergency stop, must be practiced on a regular basis.

Take the following steps if a system malfunction is detected or signaled by the system:

- Inform the duty supervisor(s).
- Evaluate the message.
- Respond to the emergency appropriately, e.g. execute an emergency stop.

Operation

Do not stay in the operating room when the product is running unless absolutely necessary and then as briefly as possible.

Keep a safe distance away from the product whenever possible. Do not touch the product unless expressly instructed to do so.

Do not inhale exhaust gases emitted by the product.

Ensure that the following requirements have been fulfilled before starting the product:

- Ear protectors are worn.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binding agent.

Operation of electrical equipment

Some components are live (high voltage) when electrical equipment is in operation.

Observe the safety instructions for these appliances.

1.4 Safety requirements for maintenance and repair work

Safety requirements before commencing maintenance and repair work

Have maintenance or repair work carried out by qualified and authorized personnel only.

Allow the product to cool down to less than 50°C (risk of explosion for oil vapors, fluids and lubricants, risk of burning).

Relieve pressure in fluid and lubricant systems and compressed-air lines which are to be opened. Use suitable collection vessels with a sufficient filling volume.

Ensure that the operating room is adequately ventilated when changing the oil or working on the fuel system.

Do not perform maintenance or repair work when the product is running unless:

- expressly instructed to do so.
- the product is running in the low load range and only for as long as necessary to complete the task.

Lock-out the product to preclude undesired starting, e.g. start interlock.

Tag-out the product with a “Do Not Start” sign in the operation room or on the control facility.

Disconnect the battery. Lock the contactor.

Close the main valve on the compressed-air system and vent the compressed-air line when pneumatic starters are fitted.

Disconnect the control facility from the product.

For starters with pinions made of copper-beryllium alloy:

- Wear a respirator mask (filter class P3). Do not blow out the interior of the flywheel housing or the starter with compressed air. Clean the interior of the flywheel housing with a class H dust extractor.
- Observe the safety data sheet.

Safety requirements during maintenance and repair work

Take special care when removing vent plugs or plug screws from the product. Hold a cloth over the screw or plug to prevent discharge of highly pressurized liquids.

Take care when draining hot fluids and lubricants (risk of burning).

Use suitable and calibrated tools only. Observe the specified tightening torques during assembly or disassembly.

Carry out work only on assemblies and/or installations which are properly secured.

Never climb up on the lines.

Keep fuel injection lines and connections clean.

Always seal connections with caps or covers if a line is removed or opened.

Fit new seals when re-installing lines.

Avoid damaging lines, particularly the fuel lines.

Ensure that all retainers and dampers are installed correctly.

Ensure that all fuel injection and pressurized oil lines are installed with enough clearance to prevent contact with other components. Never route fuel or oil lines in the vicinity of hot components.

Do not touch elastomeric seals (e.g. Viton sealing rings) with your bare hands if they have a carbonized or resinous appearance.

Observe the cooling time for components which have been heated for installation or removal (risk of burning!).

Always use suitable ladders and work platforms when working above head-height. Ensure that components or assemblies are placed on stable surfaces.

Pay particular attention to cleanliness at all times.

Safety requirements after completing maintenance and repair work

Ensure that all personnel is clear of danger zones before cranking.

Check that all access ports/apertures which have been opened to facilitate working are closed again.

Check that all safety equipment has been installed and that all tools and loose parts have been removed (especially the barring gear).

Ensure that no unattached parts have been left in/on the product (e.g. including rags and cable straps).

Welding work

Do not perform welding on the product or its attachments. Cover the product when performing welding work in the vicinity.

Before commencing welding work:

- Switch off the master power supply switch.
- Disconnect the battery.
- Disconnect electronic and genset grounds.

Do not perform maintenance or repair work on the product when welding is in progress in its vicinity. Risk of explosion or fire due to oil vapors and highly flammable process materials.

Do not use the product as a grounding terminal.

Do not route the welding cable over or near the wiring harnesses of the product. The welding current may otherwise induce an interference voltage in the wiring harnesses which could conceivably damage the electrical system.

Remove components (e.g. exhaust pipe) from the product before performing necessary welding work .

Hydraulic installation and removal

Check function and satisfactory condition of the jigs and fixtures to be used. Use only the specified jigs and fixtures for hydraulic removal/installation.

Observe the max. permissible force-on pressure specified for the installation/removal jig.

Do not attempt to bend or exert force on HP lines.

Before starting work, pay attention to the following:

- Vent the installation/removal jig, the pumps and the pipework at the relevant designated points.
- For hydraulic installation, screw on the jig with the piston retracted.
- For hydraulic removal, screw on the jig with the piston extended.

For hydraulic installation/removal jigs with central expansion pressure supply, screw the spindle into the shaft end until correct sealing has been established.

During hydraulic installation and removal, ensure that nobody is standing in the immediate vicinity of the component to be installed/removed.

Working with batteries

Observe the safety requirements of the battery manufacturer when working with batteries.

Gases released from the battery are explosive. Avoid sparks and naked flames.

Do not allow electrolyte (battery acid) to come into contact with skin or clothing.

Wear goggles and protective gloves.

Do not place tools on the battery.

Check polarity before connecting the cable to the battery. Battery polarity reversal may lead to injury by the sudden discharge of acid or bursting of the battery unit.